Dhruval Javia

djavia25@alumni.stanford.edu linkedin.com/in/dhruvaljavia/ Project Portfolio

Education

Stanford University Sept 2024 - June 2026
M.S. in Mechanical Engineering (Specialization: Energy and Transport Sciences), GPA 3.78/4.00 *Stanford, CA*
- **Relevant Coursework:** Nanoscale Heat Conduction, Convective Heat Transfer, Gas-phase Chemical Kinetics, FEA, Numerical Methods, Projection-based Reduced Order Modeling, Optical Diagnostics & Spectroscopy Lab, Li-ion cell & Fuel cell

Indian Institute of Technology, Bhubaneswar Aug 2018 – April 2022
B.Tech. in Mechanical Engineering *Bhubaneswar, Odisha*

Work Experience

Thermax Ltd. Aug 2022 – July 2024
Associate Thermal Engineer – Advent Rotational Program *Pune, Maharashtra*
- Led a cross-functional team to optimize and standardize biomass-fired thermal oil heater design, achieving around 3500 Kg weight reduction and 30 sq. m smaller footprint on average across entire series.
- Developed automated 1D iterative thermal models in Excel/VBA for the APH, Economizer, and MPA furnace; benchmarked simulations against field-calibrated design tools to achieve within 2% to 10% temperature accuracy while accelerating heater design cycles.
- Performed pressure part sizing, thermal & pressure drop analysis and weight estimation to optimize techno-commercial offering of the heater.
- Mapped manufacturing processes for boilers and heaters on the factory floor, identified bottlenecks, and proposed automation improvements; prepared detailed RLA and RCA reports based on site evaluations.

Summer Research Internship May 2021 – July 2021
IIT Madras, Project: Macro-scale Design Aspects of EV Battery Pack *Online*
- Developed a MATLAB GUI app backed by analytical battery pack models, enabling real-time slider trade-off analysis of cell parameters on overall EV range, pack weight and heat generation.
- Developed MATLAB script to simulate single-cell current profiles by integrating vehicle dynamics equations with standardized velocity profiles.

Academic Projects

Efficient Battery Thermal Management Systems For EV | *B.Tech. Thesis Project* July 2021 – April 2022
- Performed degradation studies using COMSOL Multiphysics via coupled 1-D P2D electrochemical and 2-D CFD thermal models of an air-cooled battery pack integrated with a capacity fade model.
- Optimized cell arrangement in air-cooled pack for minimum peak pack temperature and cell-to-cell temperature difference considering cell degradation due to SEI formation.
- Identified 26650 type of cells and tapered arrangements as the ideal combination to minimize volumetric heat generation and peak pack temperatures.
- Co-authored peer-reviewed journal paper in Energy Storage (Wiley, IF: 3.2): "Design of Lithium-ion battery packs for two-wheeled electric vehicles."

Gas Dynamics Simulator | *Individual Personal Summer Project* June 2025 – Aug 2025
- Built a MATLAB program from scratch using first-principles, to simulate particle dynamics and association chemical reactions in an ideal gas, implementing hash grid optimization to reduce no. of collision checks by a factor of 200.
- Developed an interactive MATLAB app with numeric and GUI-based input to improve user experience, providing time-series plots and dynamic animations as post-processing outputs for analysis.

Aerocase | *Individual Course Project, ME 203: Design and Manufacturing, Stanford University* April 2025 – June 2025
- Executed full product life-cycle including prototyping, CAD modeling, BOM generation, process sequence planning, fabrication, assembly, and functional testing under a $100 budget and 10-week timeline.
- Fabricated the case using TIG welded 0.04" thick Al 5052 aluminum sheet, A356 cast aluminum, and pine wood.

Technical Skills

Programming Languages: MATLAB, Python
Simulation: ANSYS, COMSOL, Firedrake
CAD: SolidWorks, Fusion 360, OnShape
Manufacturing: Sheet Metal Working, TIG Welding, Sand Casting, Woodworking, 3D Printing

Leadership and Achievements

Student Placement Coordinator (2022): Coordinated recruitment processes with industry and institute stakeholders.
Team Lead, e-Yantra Robotics Competition (2019-20): Collaborated with a 4-member team in the IIT Bombay robotics competition.
5th Rank, GrabCAD Shop Challenge: Ranked in the top 6% of participants globally for the design of a 3D-printable mechanical hand blender.